

Project Roadmap:

Enhancement and verification of Python-based software for image-guided surgery (Team Ctrl + Incision)

Version 1.0 – Foundation (Sprint 1A)

Theme: Set foundational standards and criteria for achievement

- Execute the current software from start to finish to fully understand its existing features.
- Record baseline runtime performance and results to establish benchmarks for comparison.
- Examine constraints and interdependencies to identify bottlenecks and opportunities for enhancement.
- Generate a system diagram illustrating the interactions between modules as they stand.
- Establish success criteria that correspond with the project's objectives of modularity, reliability, and clinical significance.
- Create a preliminary setup manual to enable future developers to efficiently replicate the environment.
- Begin reviewing the existing user interface to capture usability issues and consider areas for improvement.

Output: An operational baseline system featuring detailed documentation, quantifiable benchmarks, and an initial UI/UX assessment, setting a foundation for structured enhancements.

Version 2.0 – Modular Architecture (Sprint 1B)

Theme: Transform into a scalable, adaptable framework.

- Restructure code into clear, manageable parts to enhance organization.
- Specify the duties of each module to enhance clarity and teamwork.
- Isolate the I/O layer into an independent module for enhanced flexibility.
- Separate preprocessing into its own module to streamline workflow management.
- Implement a configuration management system to facilitate experimentation and parameter optimization.
- Incorporate unit and integration tests to ensure stability when changes are made.
- Outline the modular framework to facilitate easier onboarding for upcoming contributors.
- Refine the user interface and APIs to align with the new modular structure

Output: A flexible, expandable structure with enhanced test coverage, documentation, UI/UX aligned to the modular architecture.

Version 3.0 – 3D Reconstruction (Sprint 2A)

Theme: Construct and verify an enhanced imaging workflow.

- Examine current reconstruction methods to assess their advantages and disadvantages.
- Create unit tests to verify the results of reconstruction modules.
- Refine and enhance reconstruction code to minimize latency and boost efficiency.
- Develop an enhanced 3D reconstruction pipeline incorporated into the system.
- Verify reconstruction precision against recognized baseline implementations.
- Record algorithms, techniques, and outcomes to ensure reproducibility.
- Implement a prototype visualization of reconstructed 3D volumes in the user interface to improve usability.

Output: A trusted, verified ultrasound-to-3D reconstruction workflow, incorporated into the modular framework featuring real-time efficiency and initial UI visualization.

Version 4.0 – Integration & Validation (Sprint 2B)

Theme: Complete workflow integration and clinical verification

- Incorporate the 3D reconstruction process into the complete system workflow.
- Verify the entire system using simulated clinical scenarios to show its applicability.
- Conduct quality assurance and compliance evaluations to verify reproducibility and coding guidelines.
- Develop documentation for users that includes setup, usage, and extension guidelines.
- Review and finalize UI/UX design to make sure the created interface supports clinical usability.
- Create demonstration materials, comprising slides and a presentation, to highlight the system's clinical effects.

Output: A completely integrated and validated system ready for academic presentation and simulated clinical demonstration.

Version 4.1 – AR/Extended Vision (Exploration)

Theme: Investigation of sophisticated imaging processes

- Explore the practicality of applying AR overlays to display 3D reconstructions within surgical procedures.
- Record possible risks, advantages, and technical needs for upcoming research.

Output: A viability assessment and design documentation for expanding the system into AR-supported surgical uses.